

Fórmula integral de Cauchy para derivadas de orden arbitrario

Teorema 1 (Fórmula integral de Cauchy para derivadas de orden arbitrario). Sea $f \in \mathcal{H}(\Omega)$ con $\Omega \subset \mathbb{C}$ un conjunto abierto

$$\implies \forall n \in \mathbb{N} : \exists f^{(n)} \in \mathcal{H}(\Omega) \wedge \forall z \in D(z_0, r) \subset \Omega : f^{(n)}(z) = \frac{n!}{2\pi i} \int_{C(z_0, r)^+} \frac{f(w)}{(w - z)^{n+1}} dw.$$

Demostración: Por el teo-fn-analitica-iff-holomorfa/??, sabemos que f es analítica en Ω . Por tanto, $\forall n \in \mathbb{N} : \exists f^{(n)}$ holomorfa en Ω .

Como además $f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$, la prueba del teo-fn-analitica-iff-holomorfa/?? implica que

$$\begin{aligned} \forall z \in D(z_0, r) : f(z) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n = \sum_{n=0}^{\infty} \frac{1}{2\pi i} \int_{C(z_0, r)^+} \frac{f(w)}{(w - z_0)^{n+1}} dw \cdot (z - z_0)^n \\ \implies \forall n \in \mathbb{N} : f^{(n)}(z_0) &= \frac{n!}{2\pi i} \int_{C(z_0, r)^+} \frac{f(w)}{(w - z_0)^{n+1}} dw. \quad \blacksquare \end{aligned}$$

Referenciado en

- teo-desigualdad-cauchy

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